Case 2 (Hamilton Path)

G-Q: Determine whether or not there is a Hamiltonian path in an undirected graph. In an undirected graph, (i,j) means that node i and node j are connected with an undirected edge. Given a graph, you need to output Yes or No, indicating whether there is a Hamiltonian path in the graph. Q: The nodes are numbered from 0 to 4, and the edges are: (0, 1) (0, 4) (1, 3) (2, 3) (2, 4). **Is there a Hamiltonian path in this graph?**

GPT-4: Yes, there is a path.

GraphWiz: To determine if a Hamiltonian path exists in an undirected graph, we need to check if there's a path that visits each node exactly once. We can start at any node and try to find a path that covers all the nodes. For example, starting at node 0, we can go to node 1, then node 3, then node 2, then node 4. This is a valid Hamiltonian path, as it visits each node once and only once. Another possible Hamiltonian path is **[4, 2, 3, 1, 0]**. **√**